

Key Insights

1. High-performing games can heavily influence total sales.

A small number of high-performing games accounted for disproportionately high sales, with Wii Sports considerably outperforming other individual titles. Wii Sports is the highest-selling video game globally with a total of 82.74 million global sales. The Wii was the top platform represented in the top 10 video games sold globally, while Platform was the most represented genre.

2. Genre and platform performance varied beyond release volume.

Platform games had the highest median global sales among genres, despite having fewer releases than several other genres, suggesting that release volume alone did not correspond to stronger typical commercial performance. Meanwhile, the top-performing platform reflected by overall median global sales was Atari 2600, though it had the fewest games represented in comparison to the other platforms. Platforms with fewer than 100 games were excluded from platform-level comparisons to reduce the influence of median values that were calculated from small sample sizes.

3. Earlier years showed unusually high median sales.

Yearly median global sales in the 1980s and 1990s showed unusually high values, both in genre and platform, which coincided with a lower number of releases. This suggests that these values should be interpreted cautiously because of the smaller sample sizes. Platform games had the highest overall peak among the genres in median global sales in 1988, and Platform and Shooter games had the highest peaks in later years despite having fewer games represented than many other genres during those years.

4. Platform-genre combinations showed different performance patterns.

Median global sales for each genre and platform combination revealed that commercial performance is broadly distributed across platform-genre combinations, with several combinations showing notably high median global sales. This suggests that typical commercial performance varied based on the combination of platform and genre rather than being associated with genre or platform alone. Among platform-genre combinations with at least 10 releases, several of the combinations with the highest median global sales were on Nintendo platforms. The 3 combinations with the highest median global sales were Platform games on Game Boy, Sports games on NES, and Action games on NES.

5. Sales performance varied across platform lifecycles.

When median global sales were compared across individual platforms by years since platform release, games released during the early stages of a platform's lifecycle generally had higher median global sales, although several platforms exhibited different patterns. Median global sales for PlayStation, Wii, and Xbox 360 all peaked between 5-10 years of age, and median global sales for DS peaked at 16 years of age. The first 5 years of a platform's lifecycle were generally also the period during which the most games were released on the platform. This suggests that games released earlier in a platform's lifecycle may have had stronger median sales performance, though this relationship varied by platform.

Limitations

1. Dataset scope

This analysis is based on the historical Kaggle Video Game Sales dataset, and therefore reflects only the video game sales information represented in the dataset rather than the entire video game market.

2. Data cleaning and completeness

When cleaning the video game sales data, some regional sales values from older or incomplete records were not carried over. For games with multiple entries, records were evaluated to determine whether they represented the same game, and the most appropriate record or combination of records was retained. Additionally, some games had missing or incorrect values in their Year or Publisher columns, which were updated using information from reputable sources. When release dates differed by region, the earliest release date was selected to represent the game's initial release. This approach was consistent with the organization of the original dataset.

3. Sample size and median-based analysis

Median Global Sales was used as the primary performance measure to reduce the influence of exceptionally high-selling games. However, results in the analysis with small sample sizes should be interpreted cautiously. The visualizations provided in the analysis have been filtered to exclude comparisons based on very small sample sizes. Platforms with fewer than 100 releases were excluded from platform-level comparisons

in overall sales, and platform-genre combinations with fewer than 10 games were excluded from the primary comparison to improve the stability of median sales comparisons. These thresholds were selected by examining the results at different sample sizes and identifying points where very small samples were disproportionately represented among the highest median values.

4. Performance lifecycle methodology

Lifecycle patterns were primarily evaluated by comparing median global sales for individual platforms. The overall median by platform age was also examined, but it produced a different pattern, likely reflecting differences in the distribution of games across platforms and the influence of high-performing games at certain platform ages. Additionally, the lifecycle analysis revealed that TG16 and PC games also peaked after the first five years of the platforms' release, but it should be noted that the games present in the dataset for these two platforms were all released after the platform had already reached 5 years of age. Therefore, these platforms were not included in the lifecycle insight.

5. Causality

This analysis identifies associations between sales and video game genre, platform, release year, and platform age, but it does not establish causal relationships. Other factors such as marketing, franchise popularity, platform adoption, and competition were not controlled for.